4G and 5G in Brief

The key concepts of mobile network architecture — abridged edition for quick orientation

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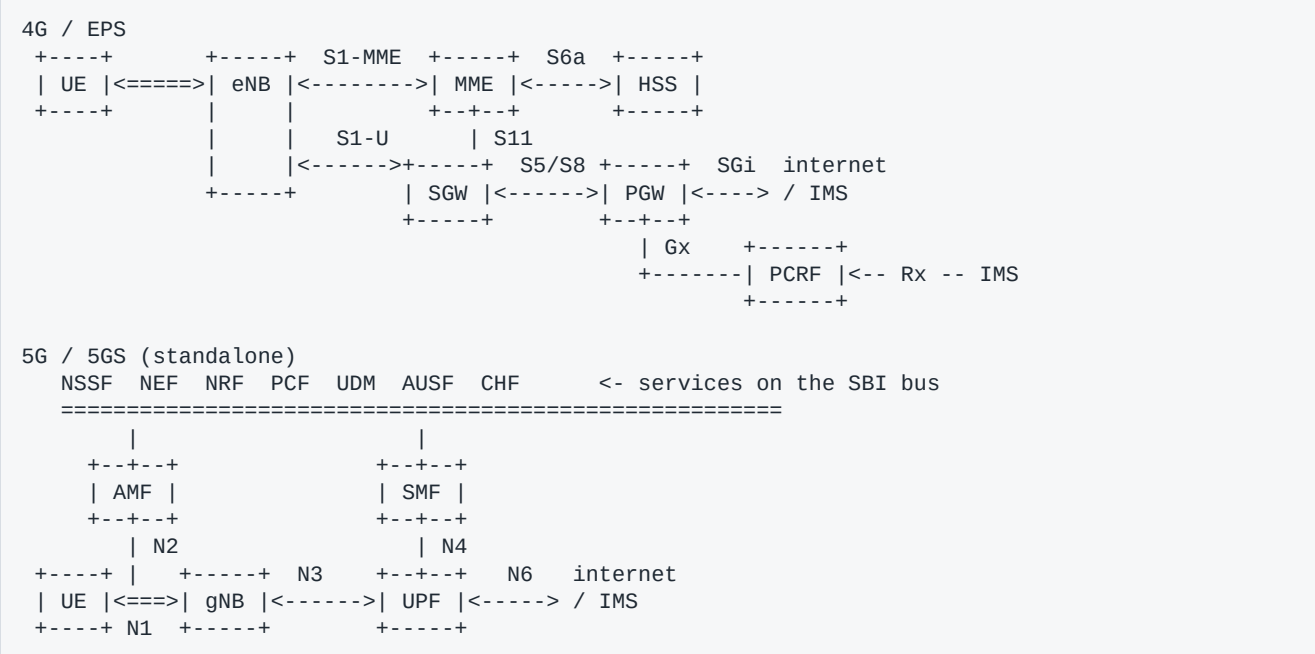
The ten things to hold in your head

1. **Both generations are all-IP.** Every service, including voice, is a packet flow. Voice is delivered by a separate system, the IMS, that sits on top of either core.
2. **Three planes.** Control (signalling: who, when, where), user (the traffic itself), management (configuration). Communications data lives on the control plane; content lives on the user plane.
3. **Radio and core are separate.** Base stations (eNB in 4G, gNB in 5G) handle the air interface; the core handles identity, sessions, policy and the route to the outside world.
4. **4G's core is a handful of boxes.** MME (control brain), HSS (subscriber database), SGW and PGW (user-plane gateways), PCRF (policy). Point-to-point interfaces with letter-number names.
5. **5G's core is a set of services.** AMF, SMF, UPF, UDM, AUSF, PCF, NRF, NSSF, NEF, CHF — cloud-native functions on an HTTP/2 bus. Control and user plane fully separated.
6. **'5G' may be 4G underneath.** Non-standalone 5G (NSA, Option 3x) uses a 5G radio on a 4G core: all records are 4G records. Standalone (SA) is the real thing.
7. **The unit of service is a session with QoS.** 4G: PDN connection and EPS bearers with a QCI each. 5G: PDU session with QoS flows with a 5QI each. Voice media is QCI/5QI 1; IMS signalling is QCI/5QI 5.
8. **Identity is layered.** Subscription (IMSI/SUPI), device (IMEI/PEI), public number (MSISDN/GPSI), IMS identities (IMPI/IMPU), temporary (GUTI), and an IP address per session. No single one is the person.
9. **Location is a cell at a signalling moment.** ECGI (4G) or NCGI (5G), captured when something happens; in idle, only the tracking/registration area. Never a track.
10. **One call, many records.** Core, IMS, policy and charging nodes each record part of it. The charging correlation identifier (ICID) is what joins them.

1. The two architectures on one page



Job	4G	5G
Base station	eNB	gNB (CU/DU split)
Mobility, registration, paging	MME	AMF
Session set-up, IP address	MME + SGW/PGW control	SMF
User-plane forwarding, NAT, usage counting	SGW, PGW	UPF
Subscriber data and authentication	HSS	UDM + UDR + AUSF
Policy and QoS decisions	PCRF	PCF
Charging	OCS / OFCS	CHF
Service discovery	(configuration)	NRF
Slicing	(none)	NSSF, S-NSSAI
External API exposure	SCEF	NEF
Wi-Fi access gateway	ePDG	N3IWF
Voice	VoLTE via IMS	VoNR via IMS, or EPS Fallback to 4G

2. Interfaces you will hear named

Name	Between	What it does
S1-MME / N2	Base station and MME / AMF	All control signalling between RAN and core; UE NAS messages ride inside
S1-U / N3	Base station and SGW / UPF	User-plane tunnels (GTP-U)
S6a / Nudm	MME / AMF and HSS / UDM	Authentication and subscription data; 'which node serves this subscriber now'
S11 / N11	MME and SGW / AMF and SMF	Create and tear down sessions
S5-S8 / N9	SGW and PGW / UPF to UPF	Gateway-to-gateway; S8 when roaming
Gx / N7	PGW and PCRF / SMF and PCF	Policy rules per flow
Rx / N5	IMS P-CSCF and PCRF / PCF	'Give this call a voice bearer' — the only place IMS touches the packet core
Sx / N4	Gateway control and user plane	PFCP rules to the user plane; where CUPS and the 5G split live
SGi / N6	PGW / UPF and the outside	The subscriber's traffic leaves the mobile network
X2 / Xn	Base station to base station	Handover without the core
N26	MME and AMF	4G-to-5G session continuity
Gm / Mw / ISC / Cx	Inside IMS	UE to P-CSCF / between CSCFs / to application servers / to HSS
LI_X1, X2, X3 / HI1, HI2, HI3	Lawful intercept	Provisioning, events, content / tasking and delivery to the authority

3. Sessions, bearers and QoS

A device with an active data connection has, in 4G, a PDN connection per APN (typically 'internet' and 'ims'), each with a default bearer and possibly dedicated bearers; in 5G, a PDU session per DNN, each carrying several QoS flows in one tunnel. The QoS class decides treatment:

QCI / 5QI	Used for	Why it matters
1	Voice media	Its set-up and release bracket a call's media; corroborates IMS timing
2	Video media	
5	IMS signalling	Exists from attach on every VoLTE/VoNR device — it means registered, not on a call
6-9	Internet traffic	Where data-session and ICR-type records come from

Because each APN/DNN has its own IP address, one device holds several addresses at once; and because most operators run carrier-grade NAT, the public address is shared by hundreds of subscribers. 'The device's IP' is only meaningful as (address, port, time).

4. The four procedures that generate records

Procedure	What happens	Records produced
Attach (4G) / Registration (5G)	UE identifies itself, is authenticated against HSS/UDM, gets a temporary identity and a tracking/registration area. In 4G a default data session is created at the same time; in 5G it is a separate step.	HSS/UDM location update; MME/AMF context with IMSI/SUPI, IMEI/PEI, cell; (4G) IP allocation
Session establishment	A PDN connection / PDU session is created to an APN/DNN; the PGW/SMF allocates an IP address; policy is fetched; tunnels are set up.	IP allocation with time; policy session; charging session start
IMS registration and call	Device registers with the IMS over the 'ims' APN/DNN; a call is a SIP INVITE through P-CSCF and S-CSCF, service logic at the TAS, a QoS bearer/flow requested over Rx/N5, media on the user plane, BYE to end.	S-CSCF session record (authoritative), P-CSCF access/location, TAS service events, IBCF interconnect, policy bearer events, IMS charging record with ICID
Mobility	Handover between cells (X2/Xn without the core, S1/N2 via it), tracking/registration area updates in idle, inter-system moves over N26, EPS Fallback for voice.	Core sees only the handovers and area updates that involve it; cell changes inside a connected session are largely invisible to the core

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A VoLTE / VoNR call, compressed
UE --INVITE--> P-CSCF --> S-CSCF --iFC--> TAS --> (I-CSCF/HSS) --> S-CSCF(term) --> P-CSCF --> callee
                 |
                 +--Rx/N5--> PCRF/PCF --> PGW/SMF --> dedicated bearer / QoS flow (QCI/5QI 1)
<====================== RTP media on the user plane, never through IMS ====================>
UE --BYE--> ... --> callee ; bearer released ; charging records closed, joined by ICID
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5. Identity in one table

Identifies	4G	5G	Note
The subscription (SIM)	IMSI	SUPI (sent concealed as SUCI)	The join key between core and IMS records
The device	IMEI	PEI	Captured at attach and IMS registration, not per call
The public number	MSISDN	GPSI	Portable and reassignable — only valid as at a time
Temporary core identity	GUTI	5G-GUTI	In traces, not disclosures
IMS credential / public identity	IMPI / IMPU	same	Several IMPUs per subscriber
The session endpoint	UE IP (per APN)	UE IP (per PDU session; may change under SSC 2/3)	Shared behind CGNAT; needs port and time
Location	ECGI (cell), TAI (area)	NCGI (cell), registration area	Different identifier spaces; cell at signalling moments only
One call across nodes	ICID	ICID	The correlation key
The other operator	IOI	IOI	Inserted at the IBCF

6. NSA, SA and the migration traps

	NSA (Option 3x)	SA (Option 2)
Radio	LTE anchor + NR secondary	NR only
Core	4G EPC	5G core
Identity	IMSI, GUTI	SUPI/SUCI, 5G-GUTI
Location id	ECGI	NCGI
Voice	VoLTE	VoNR or EPS Fallback
Records	All 4G-shaped	5G-shaped, with 4G records for fallback calls

- A '5G subscriber' on NSA produces nothing but 4G records.
- EPS Fallback splits one call across two cores: INVITE on 5G, media and release on 4G. Correlate on ICID.
- ECGI and NCGI are different identifier spaces; hold both in the cell-site reference.
- 5G UEs can change IP address mid-session (SSC modes 2/3) and hold more sessions per device (slices).
- RRC_INACTIVE in 5G hides even more mobility from the core than 4G idle did.
- Every 4G feed gains a 5G twin; reconcile volumes per feed per generation.

7. Where communications data comes from

Question	Go to
Who holds this number / subscription?	HSS / UDM and the CRM — a subscription attribute, valid as at a time
Did a call happen, between whom, when, how did it end?	S-CSCF
Where was the device when it did?	P-CSCF (P-Access-Network-Info) for IMS events; MME / AMF for core events
Which IP address did they have?	PGW / SMF allocation record
Who was behind this public IP:port then?	CGNAT translation logs — nothing else
Was the call diverted, barred, conferenced?	TAS
Which operator was the other party on?	IBCF (inter-operator identifier)
Was it 4G, 5G or Wi-Fi?	Access type field in P-Access-Network-Info; MME / AMF context
What was said / what app was used / where they were in between	Not generated. Content is user plane; OTT is encrypted; location is occasional.

The same lawful-intercept plumbing (ADMF provisioning, POIs in AMF/SMF/UPF/CSCF, MDF2 for events and MDF3 for content) carries both interception and communications data. The separation between the two is administrative — authorisation and provisioning — not physical.

8. Glossary — the short list

Term	Meaning
AMF / MME	5G / 4G mobility and registration control node
SMF / UPF	5G session control / user-plane forwarding (4G: SGW/PGW)
HSS / UDM	Subscriber database, 4G / 5G
PCRF / PCF	Policy and QoS decisions, 4G / 5G
eNB / gNB	4G / 5G base station
APN / DNN	Named data network a session connects to
Bearer / PDU session / QoS flow	The units of connectivity and QoS
QCI / 5QI	QoS class; 1 = voice media, 5 = IMS signalling
IMS, CSCF (P/I/S), TAS	Voice and messaging service core and its nodes
IMSI / SUPI, IMEI / PEI, MSISDN / GPSI	Subscription, device, public number — 4G / 5G names
ECGI / NCGI, TAI	Cell and tracking-area identifiers
ICID, IOI	Call correlation key; peer-operator identifier
NSA / SA	5G radio on a 4G core / full 5G
EPS Fallback	A 5G-originated call completed on 4G
CGNAT	Shared public IP addresses; resolution needs port and time
SBI / SBA	The HTTP/2 service bus the 5G core runs on
Slice (S-NSSAI)	A logical end-to-end network for a class of service
ADMF / POI / MDF	Lawful-intercept provisioning, capture and mediation functions